

ControlPlane

Govern the **episode**, not the response.

One questionable output shapes several downstream decisions — until an agent executes an irreversible action on a premise it fabricated four turns ago. Response-level guardrails cannot see it. ControlPlane holds it.

41/41

tainted irreversible actions held before execution

0/47

false holds on clean agent episodes

12.6 ms

median assurance overhead per request

0.04%

assurance compute vs model spend (load test)

measured — reproduced by `python -m evals.run` in the public repository

Everything Round 1 promised — measured. Then the layer nobody ships.

ROUND 1 · PROMISED

- Tier 0–3 cascade: deterministic → small models → LLM judge → human
- Inline interception during streaming
- One decision across performance, cost and responsibility
- < 60 ms added p95 (Tier-0 target) · > 90% of injected failures caught
- < 3% of model token spend · risk × reversibility decisions

ROUND 2 · MEASURED ON THE PROTOTYPE

- 12.6 ms p50 / 20.3 ms p95 added — full lite ensemble, not Tier-0 only
- 96.4–100% recall per risk category, with confidence intervals
- 0.04% of model spend (metered CPU + judge calls, load test)
- Same Tier 0–3 vocabulary, same decision set: PASS · ANNOTATE · REPAIR · ESCALATE
- Streaming claim made precise: sentence-buffered release — PII masked before the wire; released words can't be unsaid, and we log that

What changed: inline response checking is now commodity — leading platforms ship sub-200 ms guardrails. We kept ours and stopped selling it as the innovation. Round 2's differentiator is **episode-level governance: a risk budget, claim provenance and an action gate across the whole task.**

Guardrails see requests. Risk compounds across the task.

WHAT THE MARKET ALREADY SOLVED

- Response-level checks in < 200 ms, on 100% of traffic, inside your VPC — commercial platforms ship this today
- Policy mapping to EU AI Act, NIST AI RMF, ISO/IEC 42001 — baseline expectation
- Cloud providers bundle content guardrails effectively free inside existing commitments

WHAT NOBODY GOVERNS — THE BRIEF NAMES IT

“Multi-turn conversations and AI agents that take actions introduce compounding risk, where one questionable output can shape several downstream decisions.”

— Round 2 brief, Problem Track 1. A gateway sees nine model calls; it does not see nine steps of one plan.

2026: AGENTS FAILING IN THIS SHAPE

- 10 of 11 open-source coding/computer-use agents bypassed raw-string shell guards (GuardFall)
- Agent memory poisoned from a single external email (MemGhost) — one planted 'fact' steers later actions
- Defensive-review modes steered by hostile repository content (Friendly Fire)
- Only ~20% of organisations report mature AI governance (Deloitte, 2026)
- **EU AI Act Art. 14 requires effective human oversight — for an agent, the episode is the only unit at which that duty is dischargeable**

Existing guardrails ask: *is this response safe?*

ControlPlane asks: **is this task still safe, given everything that has already happened in it?**

Episode risk budget — in rupees

Every passed-but-uncertain output debits a hazard-based expected-loss meter. When the EPISODE's cumulative risk crosses the budget, it escalates — even though no single response ever crossed a block threshold.

escalated at turn 7 in the demo — ₹103,467 against a ₹100,000 budget

Claim provenance & taint

Numbers, dates, names and IDs are canonicalised — “eighty-five thousand” equals 85000. A value first appearing ungrounded in model output is tainted and tracked across every later turn. Deterministic, milliseconds, no model call.

words → digits reformatting does not launder a fabrication

Action gate on reversibility

An irreversible tool call — pay, send, delete, submit — requires a taint-clear episode, not just clean arguments. Held BEFORE execution, with the evidence chain attached for the reviewer.

41/41 tainted actions held · 0/47 false holds (measured)

Detection stays pluggable commodity — bring any guardrail vendor. The episode layer above it is the product.

Compounding risk, priced in expected loss — with honest math.

THE MATH (AND WHY NOT A NAIVE SUM)

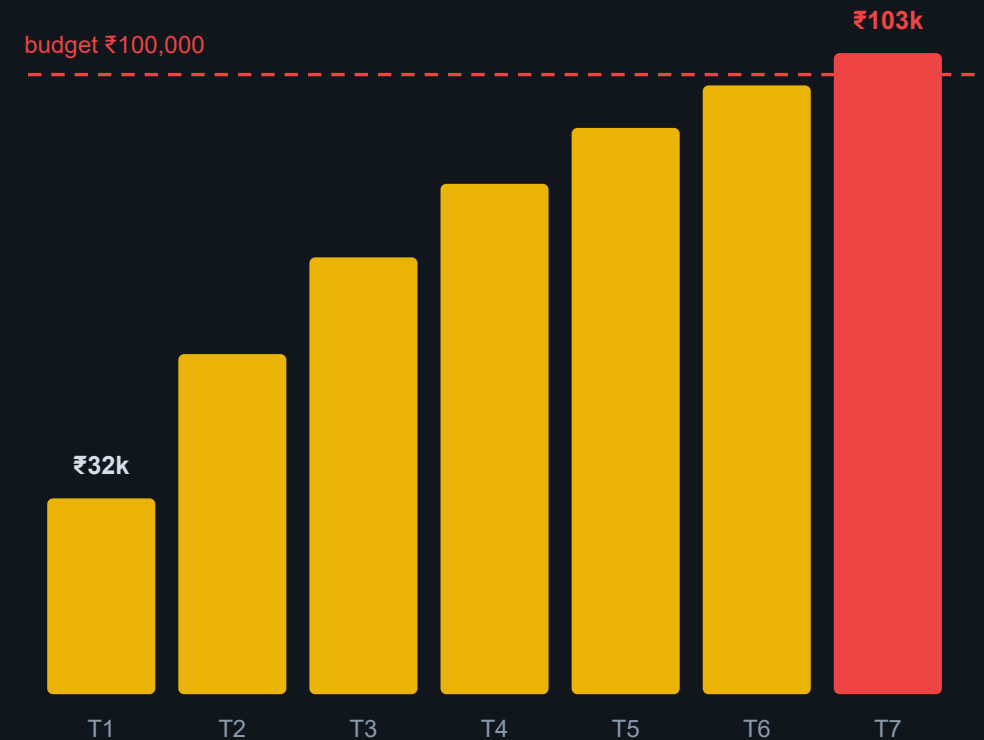
$h += -\ln(1 - p)$ per category, per turn
 expected loss = $\sum (1 - e^{-h}) \cdot \text{severity}$

Summing per-turn expected losses can exceed the worst case: 12 turns $\times$ $P=0.15 \times$ ₹50k “accrues” ₹90k against a ₹50k maximum. Hazard math is bounded, monotone, and reads as $P(\geq 1 \text{ real failure})$.

ENGINEERED AGAINST THE OBVIOUS ATTACKS

- Verbatim restatements dedupe by content hash; taint dedupes per canonical claim
- Degraded detector coverage INCREASES the debit — load cannot launder risk
- Correlated labels (fabricated person detail = grounding + privacy) debit once, at the max — both stay on the record
- Identity-scoped rolling windows — splitting sessions doesn't reset the budget

LIVE DEMO TRACE — ACT 6 (REAL OUTPUT)



“Provisional figures” from a low-trust source, turn after turn. Turns 1–6: ANNOTATE — none crosses a block threshold. Turn 7: the episode’s budget exhausts and it escalates.

Canonical values, not strings — reformatting doesn't launder a fabrication.

ONE CANONICAL VALUE

₹1,20,000 = 120000 = 1.2 lakh = one point two lakh

dates → ISO · names casefolded, honorifics stripped · IDs normalised · tolerance matching survives rounding

EVERY VALUE GETS A PROVENANCE CLASS

GROUNDING

found in a well-governed source or the user's own input

DERIVED

computable from grounded numbers — sums, GST, discounts — whitelisted, logged with its formula (models legitimately derive; tainting derivations would flood the gate)

LOW-TRUST

supported ONLY by a loosely-governed source — shared drive, forwarded email, agent memory. The brief's own data-quality split; the MemGhost defence

TAINTED

first appeared in model output with no support anywhere

WHY THIS DESIGN SURVIVES THE Q&A

- “What if the agent writes the number in words?” — canonicalisation catches it; the demo does exactly this
- Deterministic fast path on 100% of traffic: entity/number/date tracking is regex + parsing, milliseconds, zero model calls
- Full semantic claim extraction (LLM) runs only on already-elevated episodes — we say this trade-off out loud
- The same-claim-restated case dedupes; the recombined-entities case is honestly out of the lexical path's scope — it belongs to the NLI adapter and Tier-2 judge
- A claim a human verifies can be RESOLVED — the taint clears, the action re-proposes, everything ledgered

Held before execution — the shot response-level tools cannot take.

THE FLAGSHIP DEMO TRACE (DETERMINISTIC, OFFLINE, IN THE REPO)

TURN 1

“Prepare claim CLM-20391 for payout” — source says approved: ₹45,000

grounded

TURN 2

model: “payable amount comes to eighty-five thousand rupees”

TAINTED — not in any source, not derivable

TURN 3

tool call: issue_refund(claim=CLM-20391, amount=85000)

HOLD_ACTION — canonical match, words = digits

THE HARDER CASE: CLEAN ARGS, TAINTED PREMISE

Turn 1: model claims “the customer’s balance fully covers this, at ₹9,90,000” (fabricated). Turn 2: transfer_funds(amount=45000) — every argument pristine, the adjudicated amount itself.

Still held. Irreversible actions require a taint-clear EPISODE — we cannot read the model’s reasoning (API-only access), so the episode’s evidence state must be clean. Conservative by construction.

MEASURED ON 88 HELD-OUT AGENT EPISODES (OF 120 GENERATED)

41/41

tainted irreversible actions held (95% CI [91.4, 100])

0/47

false holds on clean episodes — the alert-fatigue answer

Reversible actions (lookups, drafts) pass despite taint — reversibility is the dial.

A gateway you adopt by changing one line: base_url.

INGRESS GATE

injection — direct AND inside retrieved sources · input PII · before any model call

TIERED DETECTION

Tier 0 <1 ms deterministic, 100%, never sheds · Tier 1 lexical/ONNX under the lane's latency budget · Tier 2 LLM judge, elevated episodes only → honest coverage score

RISK FUSION

isotonic calibration per detector → noisy-OR per category → correlated clusters debit once

EPISODE LEDGER

₹ hazard budget · canonical claim taint · action gate · identity windows

POLICY ENGINE

versioned, hashed, signed YAML packs · hot reload · anti-rollback · jurisdiction overlays

DECISION

PASS · ANNOTATE · REPAIR (deterministic: mask PII, hedge claims) · ESCALATE · BLOCK · HOLD_ACTION

EVIDENCE

hash-chained ledger, keyed HMAC digests (no raw text), external anchors · quarantined feedback store → retune

MEASURED TIERS (THIS HARDWARE)

Tier 0	regex/checksum/signatures	< 1 ms
Tier 1	lexical grounding, name-context, lexicons	1–5 ms
Tier 1+	ONNX NLI + toxicity (optional)	measured at warmup
Tier 2	LLM judge, elevated only	~400 ms

NER-based PII is honestly Tier 1 — not a <5 ms claim.

ONE ENGINE, THREE MODES

- GATE — block-capable; withholds tool calls (agents)
- INLINE — annotate / repair / stream (chat, copilot)
- AUDIT — shadow: observe + ledger only. This is Phase 0 and the offline log-replay assessment
- Checker faults are policy: chat fails open (annotated), regulated lanes fail closed

ControlPlane · Round 2 · 08

Three lanes, one engine — and geography is an API call, not a release.

Customer support

150 ms budget · inline · fail-open

PII repaired mid-stream, sentence-buffered release;
brand/toxicity blocked; coverage score reported

Internal copilot

1.5 s budget · inline · fail-open

grounding vs governed + loosely-governed sources;
abstains on no evidence; low-trust support annotated
with provenance

Decision-support agent

10 s budget · GATE · fail-closed

episode budget, claim taint, action gate; irreversible tools
registered in the pack; nothing unchecked delivered

THE 20-SECOND LIVE MOMENT (DEMO ACT 5)

Same episode, same content: a payout figure supported only by a forwarded email.

IN pack (DPDP posture) → transfer_funds HELD

switch jurisdiction to US — one API call, version bumps, hot-reloads

US pack (low-trust tolerated by policy) → transfer_funds PASSES

GOVERNANCE HARDENING

- Every decision records policy_version + pack_hash
- Anti-rollback: an older validly-signed pack is refused
- Unparseable pack → last-known-good keeps serving
- Operating point is an owned dial: set a benign-flag-rate target, thresholds solve from the measured sweep

MEASURED, NOT ASSERTED

Every number reproduced by `python -m evals.run` — counts and intervals included.

Risk category	Recall (caught / injected)	95% CI	False flags on benign
Grounding (unsupported assertion)	96.4% (402 / 417)	[94.2, 97.8]	0 / 1,964
Privacy (PII)	100% (132 / 132)	[97.2, 100]	0 / 2,249
Toxicity / harmful language	100% (96 / 96)	[96.2, 100]	0 / 2,285
Prompt injection (incl. indirect)	100% (68 / 68)	[94.7, 100]	0 / 2,313
Cost anomalies	100% (26 / 26)	[87.1, 100]	0 / 2,355

Test split (2,381 records) of a 3,488-record seeded synthetic dataset (~28% injected failures — labels true by construction; archetypes follow HaluEval/RAGTruth, JailbreakBench, Presidio patterns). The 15 grounding misses are entity-less recombinations — the stated limit of the deterministic path; that is what the NLI adapter and Tier-2 judge are for.

Said before you ask: these are rates on synthetic traffic with failures injected by construction. Real-traffic rates are established in the shadow phase — that is what the shadow phase is FOR. A blind hold-out slot (written by whoever didn't build the detectors) is in the repo and scored separately.

43/43

correct abstentions —
INSUFFICIENT_EVIDENCE,
never an invented score

0.065

expected calibration error
(reliability diagram on the
dashboard)

12.6 ms

p50 added per request (p95 20.3
ms, ~250-token responses)

108 req/s

single instance at saturation · 0
errors · stateless scale-out

0.04%

assurance vs model spend —
load test; 0.2% on the failure-
rich eval mix

>96%

of traffic checked with no LLM
call at all

docker compose up — fully offline, no keys, no model downloads.

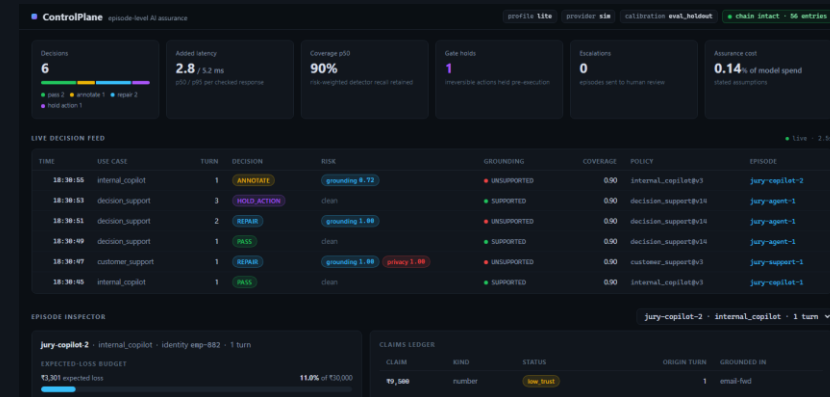
THE 10-ACT SCRIPTED DEMO (DETERMINISTIC)

- PII masked mid-stream — sentence-buffered release
- Honest abstention · governed-vs-loosely-governed source split
- Words-vs-digits taint catch → issue_refund HELD before execution
- Audit-mode side-by-side: what “off” looks like — the payout executes
- IN → US jurisdiction flip reverses the gate outcome, no restart
- Budget exhaustion at turn 7 — compounding, not thresholds
- Two-person override (self-approval refused) + overturn-rate watch
- Checker crash: chat fails open, regulated lane fails closed
- Indirect injection inside a poisoned document — blocked at ingress
- Hash-chain verification, telemetry, LLM-vs-non-LLM split, cost meter

```
git clone <repo> && cd controlplane
docker compose up --build           # gateway + dashboard :8080
python -m demo.run_demo             # the 10 acts, offline
python -m evals.run                 # reproduce every number
```

OPERATOR CONSOLE (REACT)

live decision feed · episode inspector with ₹ budget gauge, hazard bars, claims & gate events · operating-point console over the measured sweep · eval scoreboard with reliability diagram · policy-pack cards with live jurisdiction switch · override queue



REPO DISCIPLINE

- 24 tests · seeded dataset · every deck number scripted
- Apache-2.0 · NOTICE with model/benchmark attribution
- Replay-first: the stage demo survives venue Wi-Fi

They are excellent at responses. The episode is unclaimed.

Player	What they do well (credited)	Where the ceiling is
Fiddler · Galileo · Arthur	in-VPC small-model checks on 100% of traffic; sub-200 ms guardrails; session observability	requests, not reasoning — no cumulative ₹ risk, no provenance-gated actions. Observability after the fact ≠ enforcement before the irreversible call
Hyperscaler guardrails (Bedrock, Azure)	content safety effectively free inside existing cloud commitments — the real pricing threat	per-response, single-cloud; no cross-vendor evidence ledger, no episode state
AI gateways (LiteLLM, Portkey)	routing, quotas, guardrail hooks at the proxy	plumbing, not governance — no calibrated decisions, no audit-grade ledger

Positioning: the neutral episode & evidence layer ABOVE whatever detectors the client already gets — including free ones. Detectors are pluggable adapters; commoditised detection makes the governance layer MORE valuable, not less. “Galileo ships session metrics” → session observability after the fact vs episode enforcement before the irreversible action — the demo's held tool call is the proof. And an episode ledger is a fast-follow for them only if they abandon per-response pricing and architecture — our moat is the unit of account.

Prepared for the obvious probe — “Why wouldn't the client just buy them?” They can. Bring-your-own-detector is a feature: the assessment, the signed severity table, the episode ledger and the operating methodology are what Accenture sells on top.

Accenture's own motion: assess → deploy → run. Priced on the unit we govern.

Economic buyer: CIO / Head of AI Platform — with the CRO as co-sponsor whose regulatory pressure creates urgency. Engineering, FinOps and operations are stakeholders to neutralise (latency, cost, alert fatigue), not signatures to collect.

LAND

Assurance Assessment — 4–6 weeks, fixed fee

Runs in AUDIT mode over exported LLM logs — zero production footprint, no InfoSec cliff before week 1. Deliverables: the episode risk report no incumbent can produce, and the client-signed severity table (their risk office signs the ₹ numbers — we never invent them).

EXPAND

In-tenancy deployment

Priced per governed use case / agent class per month, tiered by risk class and jurisdiction count. Request metering is a fair-use throttle, not the headline. Assurance COGS $\approx$ 0.04% of model spend is the margin story, never the price anchor.

RUN

Managed assurance service

The escalation triage bench (L1/L2/L3), quarterly policy-pack updates, drift recalibration against a frozen canary set. The human-review labour IS the recurring service line — priced in, not hidden from the ROI.

Worked example at the brief's own volume (2.1M interactions/yr, 3 use cases): $3 \times ₹5\text{L platform} + ₹10\text{L managed service (incl. the triage bench)} \approx ₹25\text{L/yr per client}$ — against ₹1.2Cr+ modelled avoided loss (next slide). Why Accenture builds it: this productises what the Responsible AI practice already does manually — thousands of GenAI engagements are the install base.

A floor you can defend, a band you can believe — assumptions on the slide.

FLOOR CASE — REMEDIATION ONLY, ZERO REGULATORY TERM

Volume (brief parameter)	40,000 interactions/week $\approx$ 2.1M/yr
Material-failure rate (assumed $\rightarrow$ measured in shadow)	1.0%
Catch rate (measured on benchmark)	95%
Cost per failure reaching a user (remediation + concession)	₹600
Avoided loss floor	$\approx$ ₹1.2 Cr / yr
All-in assurance cost: $3 \times ₹5\text{L platform} + ₹10\text{L managed service}$ — which INCLUDES the triage bench (0.4% escalation rate $\times$ 2.1M = $8,400/\text{yr} \times 10 \text{ min} \times ₹500/\text{hr} \approx ₹7\text{L}$)	$\approx$ ₹25L / yr
ROI floor	$\approx 4.8\times$ before any regulatory term

WHY THIS SURVIVES A CFO

- Every input is on the slide — attack the assumption, not the method
- The failure rate is not ours to claim: Phase 0 shadow mode measures it on the client's own traffic
- The severity table is not ours to invent: the client's risk office signs it during the assessment — that signature makes the episode budget THEIR number
- Triage labour sits inside the denominator — the ROI is net of the humans
- The assurance meter and the episode budget are the same ₹ machinery: the ROI dashboard is the product, not a spreadsheet

SCENARIO BAND — SHOWN SEPARATELY, NEVER BLENDED IN

Probability-weighted regulatory and brand exposure (DPDP Act obligations for personal-data safeguards; incident disclosure; goodwill) adds a modelled ₹30L–₹1.5Cr expected term. We never anchor on the statutory ceiling — multiplying a tiny probability by a maximum penalty is fear math a CFO discounts on sight.

The episode layer exists from day zero. Phases gate enforcement authority, not capability.

0 · OBSERVE

wk 0–6

audit mode on exported logs / shadow: full episode ledger runs, zero enforcement.
Outputs: FP/FN baseline, benign-episode budget calibration, episode risk report

1 · ENFORCE RESPONSES

wk 6–14

one use case, conservative operating point set on the console; overrides start feeding retune

2 · ENFORCE EPISODES & ACTIONS

mo 4–8

budgets + action gate get block/hold authority on the regulated lane; two-person override live

3 · ESTATE

mo 8–18

all use cases, jurisdiction packs, detector-adaptor marketplace (bring your own vendor), cross-estate identity windows

False-positive fatigue	shadow first · operating point is the client's dial · 0/47 false holds measured · overturn-rate watch
Latency regression	budget scheduler with per-detector timeouts · honest coverage score · measured 12.6/20.3 ms
Checker fails or is attacked	fail-open/closed per lane · Tier 0 never sheds · ingress gate · GuardFall-informed red-teaming
Detector drift / vendor change	weekly recalibration vs frozen canary set · model fingerprint on every decision
Ledger becomes a privacy liability	keyed HMAC digests, no raw text · split quarantined feedback store · retention policy
Fast-follow by incumbents	moat is the unit of account + the signed severity methodology + Accenture's delivery install base

Every brief requirement has a mechanism. Every boundary is ours, said first.

Different risk/latency budgets per use case	policy packs + budget scheduler + coverage score
Overlapping risk categories	multi-label vector; correlated clusters — two labels, one debit (132/132)
No reliable real-time ground truth	unsupported-assertion detection + first-class abstention (43/43)
Over- vs under-flagging tradeoff	operating-point console over the measured sweep; 0 false flags at chosen point
Multi-turn & agent compounding risk	₹ episode budget · claim taint · action gate (41/41, 0/47)
Evolving regulation by geography	jurisdiction overlays, hot-reload, versioned + hashed + anti-rollback
API-only model access	pure I/O layer; opportunistic logprobs recorded in the fingerprint, never faked
Governance & audit trail	hash-chained HMAC ledger, external anchors, two-person overrides
Feedback loops	quarantined override store → threshold retune → before/after on the dashboard
Metrics for a sceptical stakeholder	counts + Wilson CIs + ECE + reliability diagram + reproduction scripts

WHAT IT DELIBERATELY DOES NOT DO

- Verify truth — we detect unsupported assertion and abstain
- Inspect model internals — API-only is the brief's constraint
- Train custom detectors — adapters are commodity; governance is the IP
- Claim benchmarked bias detection — the heuristic can annotate, never block; said out loud

The one-line close

Every AI programme already checks its responses. Nobody governs its tasks. ControlPlane is the control point at the only moment that matters — before the irreversible action.